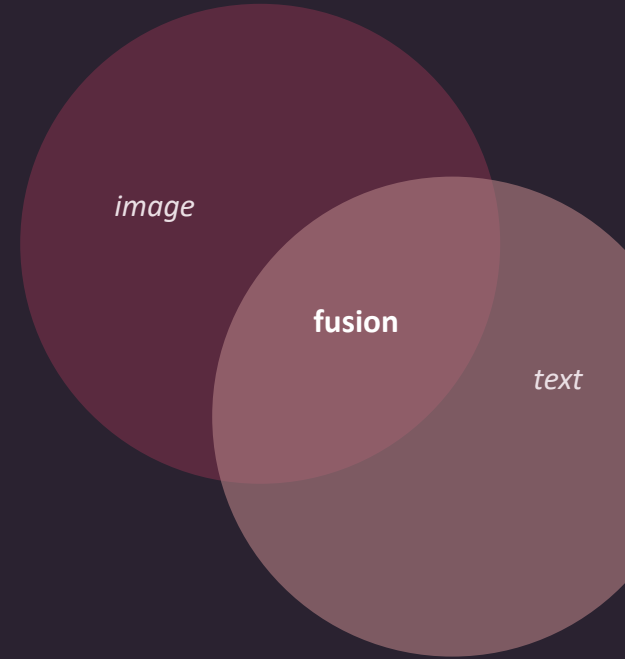


DEEP LEARNING · COURSE PROJECT

Multi-Modal Deep Learning for Fashion Product Classification

Image + text → product subcategory



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MOTIVATION

The central question

Does combining image and text beat using either modality alone?

Input: product image + productDisplayName

Output: subCategory (top-10 classes)

1

Image-only

CNN branch alone

2

Text-only

RNN branch alone

3

Fusion

image + text combined

The three-way comparison — on one identical pipeline — is the whole point of the project.



DATA

The dataset

- **Kaggle Fashion Product Images (small)**
 - ships product images, labels, and a text field
- Target: subCategory — top-10 most frequent classes
- Text modality: productDisplayName
- Stratified 70 / 15 / 15 split, fixed seed
 - identical split + vocabulary for all three models

Why this target?

✗ masterCategory

too easy — fusion would be invisible

✗ articleType

140+ classes — too hard to train fairly

✓ top-10 subCategory

the sweet spot — fair to compare



PIPELINE

Architecture at a glance

IMAGE BRANCH — CNN



TEXT BRANCH — RNN

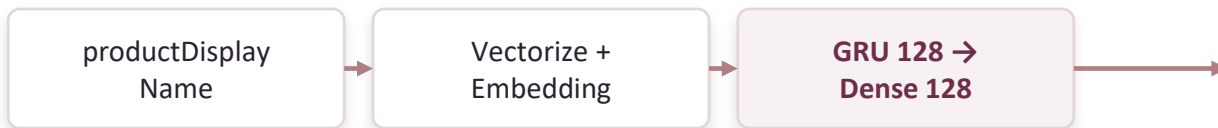


Image-only and text-only baselines reuse these identical encoders → a fair test. Built with the Keras functional API (TensorFlow), runs on a free Colab GPU.



COMPONENT 1

Image branch — CNN

- **Convolution** — shared local kernels → translation-equivariant feature maps
- **Activation** — ReLU in our head; EfficientNet uses the smoother Swish
- **Pooling** — Global Average Pooling → one value per channel, no parameters
- **Backbone** — EfficientNetB0, ImageNet weights, frozen

Why freeze the backbone?

- transfer learning — ImageNet features reuse well
- less overfitting on a modest dataset
- cheaper to train
- BatchNorm stays in inference mode → stable

Caveat: images are ~60×80 px, upscaled to 224×224 → a visual ceiling on this branch.



COMPONENT 2

Text branch — embedding + GRU

- **Trainable embedding**
 - learned end-to-end; no GloVe / BERT — the text is short and domain-specific
- **GRU recurrence — gates control memory across the sequence**
 - update gate: carry the old state vs. write the new one
 - reset gate: how much past history to forget
- **Final hidden state → Dense(128) = text encoding**

GRU gating

$$z_t = \sigma(W_z e_t + U_z h_{t-1} + b_z) \quad (\text{update gate})$$

$$r_t = \sigma(W_r e_t + U_r h_{t-1} + b_r) \quad (\text{reset gate})$$

$$\tilde{h}_t = \tanh(W_h e_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (\text{candidate})$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (\text{output})$$

The $(1 - z_t)$ carry path lets gradients flow across many steps — no vanishing gradient.



Fusion — why concatenation + FC works

Concatenate the two encodings, then a dense layer with ReLU + dropout:

$$\hat{y} = \text{softmax}(W_o \cdot \text{ReLU}(W_{\text{img}} \cdot a_{\text{img}} + W_{\text{txt}} \cdot a_{\text{txt}} + b) + b_o)$$

- **Cross-modal conjunctions** — a hidden unit fires only when a visual feature and a textual feature co-occur
- **Tie-breaking** — when the image is ambiguous the text decides — and vice-versa
- **Regularisation** — Dropout 0.3 on the only high-capacity trainable part of the model



TRAINING

Loss & optimisation

Softmax + cross-entropy

$$\hat{y}_k = \text{softmax}(o)_k = \frac{e^{o_k}}{\sum_{j=1}^K e^{o_j}}$$

o_k : logit for class k
 $\hat{y}_k$: predicted probability of class k
All probabilities sum to 1.

$$L = - \sum_i \sum_k y_{ik} \log(\hat{y}_{ik})$$

Cross-Entropy Loss

$$\frac{\partial L}{\partial o_k} = \hat{y}_k - 1[k = y_i]$$

Gradient

Adam — adaptive steps

momentum (1st moment) + RMS (2nd moment)

$$\theta_t = \theta_{t-1} - \eta \cdot \hat{m}_t / (\hat{v}_t + \varepsilon)$$

Why Adam here — the model mixes a frozen CNN, a trainable embedding and a recurrent net, with very different gradient scales — per-parameter rates handle this without manual tuning.

- Early stopping on val loss (patience 3, restore best weights)
- Seed = 42 everywhere → reproducible runs



CONFIGURATION

Experimental setup

Setting	Value	Setting	Value
Optimiser	Adam, lr 1e-3	Image	224×224, EffNetB0 frozen
Loss	sparse cat. cross-entropy	Vocab / seq	10k / length 20
Batch / epochs	32 / ≤15 (early stop)	Embed / GRU	128 / 128
Split	70/15/15 stratified	Fusion FC	256, dropout 0.3
Evaluation	held-out test (not k-fold)	Seed	42

All hyperparameters centralised in `src/config.py`. `k`-fold skipped deliberately — the variance estimate isn't worth $k\times$ the compute on a frozen-backbone model.



RESULTS

Results — the three-way comparison

Model	Accuracy	Macro-F1
Image-only	0.9827	0.9727
Text-only	0.9970	0.9949
Fusion (multi-modal)	0.9947	0.9905

HEADLINE

Text saturates the task.

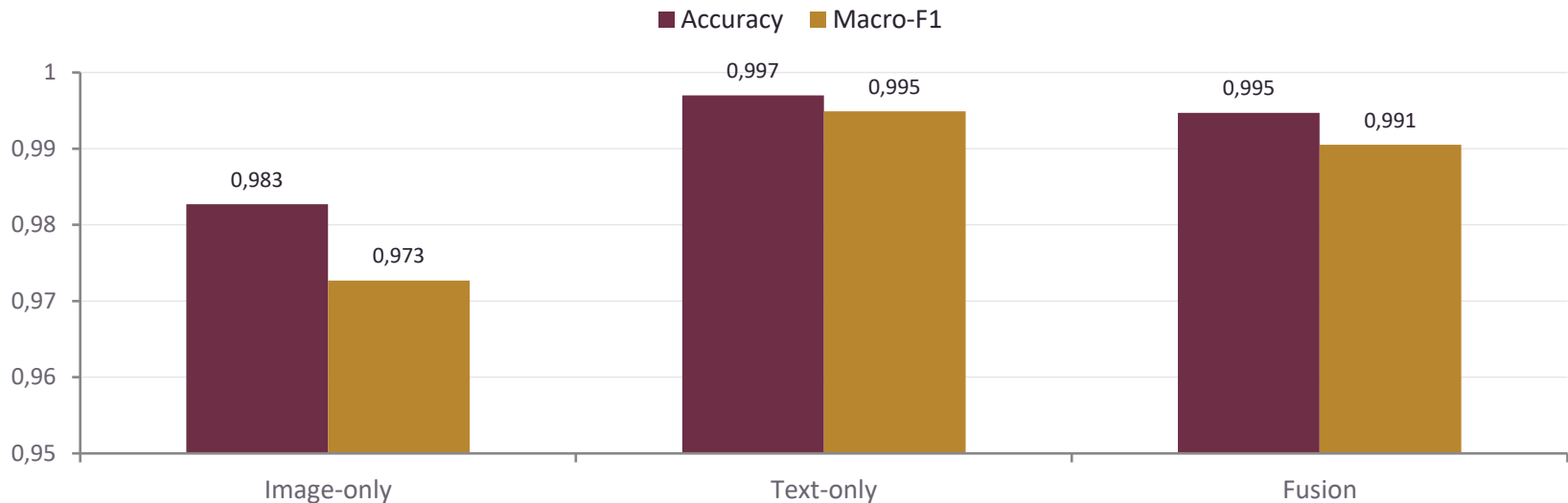
Fusion ties the best modality ($\Delta F1 -0.004$, within noise) and both beat image-only (+0.018).

We report macro-F1, not plain accuracy: the top-10 classes are imbalanced, and macro-F1 weights every class equally.



RESULTS

Results — visualised



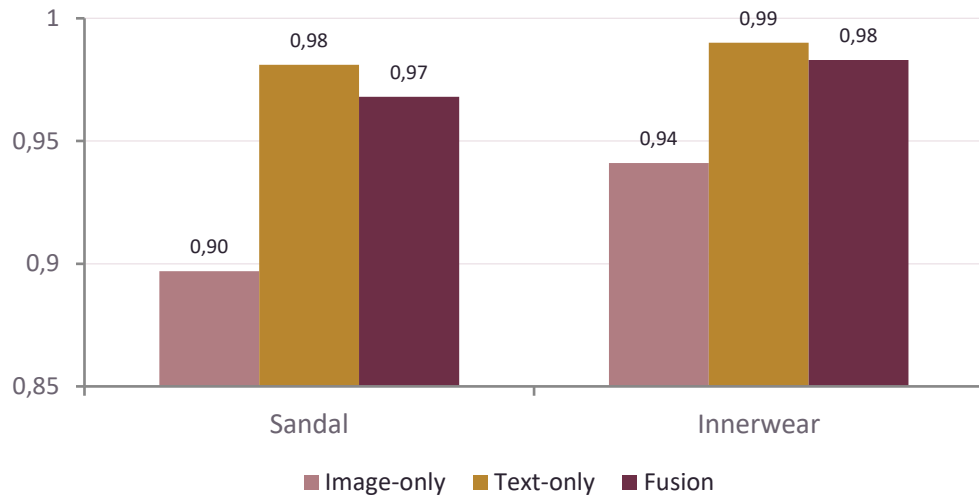
Axis zoomed to 0.95–1.00. Text-only and fusion are visually indistinguishable; image-only trails on both — fusion matches the dominant modality rather than beating it.



PER-CLASS

Where fusion actually helps

F1 on the image branch's two weakest classes



Fusion repairs the image branch's worst confusions by leaning on text:

Sandal 0.897 → 0.968 (was confused with Shoes)

Innerwear 0.941 → 0.983

Text-only is already near-perfect, so fusion's gains show up per-class — not in the overall average.



DISCUSSION

Limitations & honest scope

1

Low-resolution images

~60×80 px upscaled to 224×224 → a hard ceiling on the visual branch

2

Very descriptive text

productDisplayName nearly names the class → little fusion headroom

3

Single held-out split

no k-fold — a deliberate compute trade-off, some variance remains

4

Frozen backbone

the CNN can't specialise to fashion textures (stage-2 fine-tune = stretch)

TAKEAWAYS

- Built the full mandated pipeline — CNN + RNN + fusion, end to end
- Derived the maths of every component (conv/pool/activation, GRU gates, cross-entropy + Adam, fusion)
- Fair comparison — identical pipeline, shared encoders, no leakage
- Reproducible — fixed seed, persisted vocab, one-click Colab notebook

Central result: text saturates this task → fusion ties it and both beat image; fusion's gains are per-class, recovering the image branch's hardest categories.

Thank you for listening! — Q&A